

# Note

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## 1 February 26

### 1.1 Sample Preparation [1]

#### 1.1.1 For $W_L^\pm W_L^\pm$

```
generate p p > w+{0} w+{0} j j QCD=0, w+ > l+ vl
add process p p > w-{0} w-{0} j j QCD=0, w- > l- vl~
output WW_LL
launch
shower = pythia8
detector = Delphes (R=0.4)
```

#### 1.1.2 For $W_L^\pm W_T^\pm$

```
generate p p > w+{0} w+{T} j j QCD=0, w+ > l+ vl
add process p p > w-{0} w-{T} j j QCD=0, w- > l- vl~
output WW_LT
launch
shower = pythia8
detector = Delphes (R=0.4)
```

#### 1.1.3 For $W_T^\pm W_T^\pm$

```
generate p p > w+{T} w+{T} j j QCD=0, w+ > l+ vl
add process p p > w-{T} w-{T} j j QCD=0, w- > l- vl~
output WW_TT
launch
shower = pythia8
detector = Delphes (R=0.4)
```

#### 1.1.4 For Unpolarized $W^\pm W^\pm$

```
generate p p > w+ w+ j j QCD=0, w+ > l+ vl
add process p p > w- w- j j QCD=0, w- > l- vl~
output WW_UN
launch
shower = pythia8
detector = Delphes (R=0.4)
```

## 1.2 Restrictions for Candidates

### 1.2.1 Electrons

Electrons are required to satisfy the “Tight” likelihood-based identification criteria.

- $p_T > 27$  GeV
- $|\eta| < 2.47$  GeV
- Excluding the calorimeter transition region  $1.37 < |\eta| < 1.52$

```
ele_mask = (eles.pt > 27) & (abs(eles.eta) < 2.47) & ~((abs(eles.eta) > 1.37) &  
(abs(eles.eta) < 1.52))
```

### 1.2.2 Muons

Muons are required to satisfy the “medium” identification selection.

- $p_T > 27$  GeV
- $|\eta| < 2.5$

```
mu_mask = (muons.pt > 27) & (abs(muons.eta) < 2.5)
```

### 1.2.3 Jets

Jets are reconstructed using the anti- $k_t$  algorithm with a radius parameter of  $R = 0.4$ , using the particle-flow object.

- At least two jets with  $|\eta| < 4.5$  are required to be present in each event
- $p_T > 65$  GeV for leading jets (written in the paragraph for election of the leading leptons later.)
- $p_T > 35$  GeV for subleading jets

```
jet_mask = (jets.pt > 35) & (abs(jets.eta) < 4.5)
```

## 1.3 Selection Cuts

The event selection in the signal region (SR) requires:

1. A same-sign lepton pair with  $m_{\ell\ell} > 20$  GeV

```
padded_leps = ak.pad_none(all_leps, 2)  
l1, l2 = padded_leps[:, 0], padded_leps[:, 1]
```

```
mask1 = (ak.num(all_leps) >= 2) & ak.fill_none((l1.q == l2.q) & ((l1 + l2).mass  
> 20), False)  
n1 = ak.sum(mask1)
```

2.  $E_T^{\text{miss}} > 30$  GeV

```
met_pt = ak.firsts(data.MissingET.MET)
mask2 = mask1 & ak.fill_none((met_pt > 30), False)
n2 = ak.sum(mask2)
```

3. VBS topology: The two highest- $p_T$  jets must satisfy  $m_{jj} > 500$  GeV with  $|\Delta y_{jj}| > 2.0$

```
padded_jets = ak.pad_none(sel_jets, 2)
j1, j2 = padded_jets[:, 0], padded_jets[:, 1]

vbs_kinematics = (j1.pt > 65) & (j2.pt > 35) & ((j1 + j2).mass > 500) & (abs(j1.
rapidity - j2.rapidity) > 2.0)
mask3 = mask2 & (ak.num(sel_jets) >= 2) & ak.fill_none(vbs_kinematics, False)
n3 = ak.sum(mask3)
```

| Cut        | LL    | pass rate | LT    | pass rate | TT    | pass rate | Unpol. | pass rate |
|------------|-------|-----------|-------|-----------|-------|-----------|--------|-----------|
| Total      | 50000 | 1.00      | 50000 | 1.00      | 50000 | 1.00      | 50000  | 1.00      |
| Lepton cut | 12534 | 0.25      | 12453 | 0.25      | 12814 | 0.26      | 12491  | 0.25      |
| MET cut    | 10327 | 0.21      | 10614 | 0.21      | 11156 | 0.22      | 10745  | 0.21      |
| Jet cut    | 4403  | 0.09      | 5254  | 0.11      | 5477  | 0.11      | 5252   | 0.11      |

Table 1: Event Pass Rates across different polarization states.

## 1.4 Kinematic Distributions

### 1.4.1 Definition and Physical Meaning

#### 1. Transverse Mass $m_T$

In the lab frame, we only know the transverse components of the neutrinos (the Missing Transverse Energy, or MET). Because the longitudinal momentum ( $p_z$ ) of the neutrinos is unknown, we cannot calculate a true invariant mass. Instead, we project everything onto the 2D transverse plane ( $x - y$ ) to calculate the Cluster Transverse Mass ( $m_T$ ).

The Mathematics: To calculate the  $m_T$  of the  $WW$  system, we treat the two visible leptons as a single combined “cluster” ( $\ell\ell$ ) and the MET as the invisible component. Dilepton Transverse Energy ( $E_T^{\ell\ell}$ ): This combines the transverse momentum and the invariant mass of the dilepton system.

$$E_T^{\ell\ell} = \sqrt{|\vec{p}_T^{\ell\ell}|^2 + M_{\ell\ell}^2}, \quad (1)$$

Missing Transverse Energy ( $E_T^{\text{miss}}$ ): This is simply the magnitude of the MET vector.

$$E_T^{\text{miss}} = |\vec{p}_T^{\text{miss}}|, \quad (2)$$

$WW$  Transverse Mass ( $m_T^{WW}$ ): We subtract the squared vector sum of the transverse momenta from the squared scalar sum of the transverse energies.

$$m_T^{WW} = \sqrt{(E_T^{\ell\ell} + E_T^{\text{miss}})^2 - |\vec{p}_T^{\ell\ell} + \vec{p}_T^{\text{miss}}|^2}. \quad (3)$$

## 2. Difference in Azimuthal Angle $\Delta\phi_{jj}$

$$\phi = \tan^{-1} \left( \frac{p_y}{p_x} \right) \rightarrow \Delta\phi_{jj} = |\phi_{j1} - \phi_{j2}|, \quad (4)$$

Measures the angle between the two tagging jets in the transverse plane.

## 3. Difference in Pseudorapidity $\Delta\eta_{\ell\ell}$

$$\eta = \frac{1}{2} \ln \left( \frac{|p| + p_z}{|p| - p_z} \right) \rightarrow \Delta\eta_{\ell\ell} = |\eta_{\ell1} - \eta_{\ell2}|, \quad (5)$$

Measures how far apart the two charged leptons are along the longitudinal axis.

## 1.4.2 Results

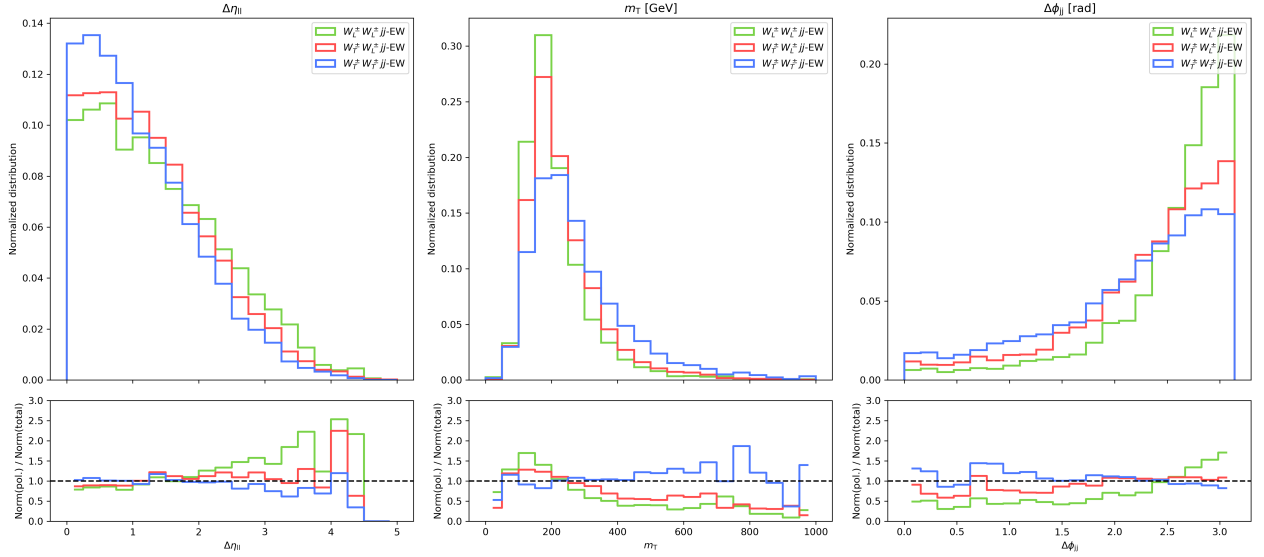


Figure 1: Distribution in the signal region for the difference in pseudorapidity between leading and subleading leptons, the transverse mass, and the difference in azimuthal angle between leading and subleading jets.

For the bottom panel, the ratio is calculated by comparing the normalized distribution of a polarized sample (LL, LT, or TT) against the normalized distribution of the unpolarized (total) sample. The ratio is derived for the plotting:

Before calculating the ratio, the code ensures both the polarized and unpolarized samples are on the same scale by normalizing them. Instead of using the raw number of events, it treats each event as a fraction of the total sample size:

$$w_i = \frac{1}{N_{\text{events}}}, \quad (6)$$

where  $N_{\text{events}}$  is the number of events that passed all cuts for that specific sample.

For each variable, the data is divided into  $k$  bins. Let  $H_p(b)$  be the normalized count for a polarized sample in bin  $b$ , and  $H_u(b)$  be the normalized count for the unpolarized sample:

$$H(b) = \sum_{x_i \in \text{bin } b} w_i, \quad (7)$$

The ratio plotted in the bottom panel represents how much a specific polarization deviates from the average unpolarized behavior in a specific kinematic region:

$$R(b) = \frac{\text{Norm(Polarized)}_b}{\text{Norm(Unpolarized)}_b} = \frac{H_p(b)}{H_u(b)}, \quad (8)$$

- If  $R(b) > 1$ : The polarized state is more likely to produce events in that specific parameter range than the total population.
- If  $R(b) < 1$ : The polarized state is suppressed in that range.

## 2 March 1st

### 2.1 Additional

To address the low selection efficiency observed in the initial MadGraph results<sup>1</sup>, I implemented additional phase-space cuts directly in the run card and increased the number of events generated. This refinement ensures a higher yield of relevant events and optimizes computational resources.

I adjusted the following parameters in the run\_card. Including restrictions for the transverse momentum of leptons and jets, missing transverse mass, pseudorapidity for leptons and jets, invariant mass for lepton pairs and jet pairs, and the transverse momentum for leading and subleading jets.

```

27.0 = ptl
35.0 = ptj
20.0 = misset
2.5 = etal
4.5 = etaj
20.0 = mml1
500.0 = mmjj
65.0 = ptj1min
35.0 = ptj2min

```

| Event Pass Rates<br>Cut | LL       |      | LT       |      | TT       |      | Unpol      |      |
|-------------------------|----------|------|----------|------|----------|------|------------|------|
|                         | Count    | Rate | Count    | Rate | Count    | Rate | Count      | Rate |
| Total                   | 100000   | 1.00 | 100000   | 1.00 | 100000   | 1.00 | 100000     | 1.00 |
| Lepton cut              | 26793    | 0.27 | 27563    | 0.28 | 30021    | 0.30 | 28388      | 0.28 |
| MET cut                 | 23816    | 0.24 | 25036    | 0.25 | 27755    | 0.28 | 25979      | 0.26 |
| Jet cut                 | 19002    | 0.19 | 20602    | 0.21 | 22352    | 0.22 | 21116      | 0.21 |
| Cross Sections          | LL (pb)  |      | LT (pb)  |      | TT (pb)  |      | Unpol (pb) |      |
|                         |          |      |          |      |          |      |            |      |
| Total                   | 0.000505 |      | 0.002843 |      | 0.004288 |      | 0.007653   |      |
| Lepton cut              | 0.000135 |      | 0.000784 |      | 0.001287 |      | 0.002173   |      |
| MET cut                 | 0.000120 |      | 0.000712 |      | 0.001190 |      | 0.001988   |      |
| Jet cut                 | 0.000096 |      | 0.000586 |      | 0.000959 |      | 0.001616   |      |

Table 2: Event pass rates and cross sections after pre-cut in MadGraph for LL, LT, TT, and Unpolarized samples.

In table 2, we can see that applying pre-cutting in MadGraph improves the event throughput, which increases the efficiency of data utilization. Furthermore, the change in cross-section after each cut is exactly the ratio times the original cross-section.

## 2.2 Transformation from $pp$ -cmf to $WW$ -cmf

Since the kinematic distribution in the paper[1] is under  $WW$ -cmf, we should consider to transform our original generated events in MadGraph from  $pp$ -cmf to  $WW$ -cmf.

### 2.2.1 Work Flow

To transform the cmf of the events, we have to first extract the 4-vector from the original events including leptons, neutrinos, and  $W$  bosons, where  $W = l + \nu_l$ . Then, we calculate the  $WW$  system kinematic by:

$$P_{WW} = P_{W+} + P_{W-}, \quad (9)$$

where  $P$  is the 4-vector.

To move into the  $WW$ -cmf, we must apply a Lorentz transformation to boost our target particles in the opposite direction of the  $WW$  system's motion. Therefore, we boost using  $-\vec{\beta}$ .

$$\vec{\beta}_{WW} = \frac{\vec{p}_{WW}}{E_{WW}}, \quad (10)$$

where  $\vec{p}_{WW}$  equals  $(p_x, p_y, p_z)$ .

Since we only know  $(E_\ell, p_{\ell,x}, p_{\ell,y}, p_{\ell,z})$  and  $(p_{\nu,x}, p_{\nu,y})$ , we have to calculate the  $p_{\nu,z}$  and  $E_\nu$ . By using the  $W$ -mass constraint, which assumes the  $W$  boson is perfectly on-shell,

$$(p_\ell + p_\nu)^2 = m_W^2, \quad (11)$$

where  $m_W \approx 80.4$  GeV.

By expanding equation 11 using the known lepton kinematics and the measured transverse missing energy, we can end up with a standard quadratic equation for the neutrino's longitudinal momentum and then figure out the  $p_{\nu,z}$  and  $E_\nu$ . However, the transverse part of the momentum is non zero, therefore, the boost is not a purely longitudinal boost, but a 3D boost.

### 2.2.2 Expectations after the Transformation

- $\Delta\eta_{\ell\ell}$ :

$$\Delta\eta' \neq \Delta\eta, \quad (12)$$

The difference in pseudorapidity should change on an event-by-event basis. Because a 3D boost mixes the momentum components, there is no single constant shift applied to the leptons. The  $\Delta\eta$  distribution in the boosted frame should look smeared or broaden compared to the lab frame.

- $m_T$ :

$$m_{T,pp-cmf} \rightarrow M_{WW,WW-cmf}, \quad (13)$$

Since we calculated the longitudinal momentum for neutrino, we can then figure out the invariant mass, instead of the transverse mass. While the total momentum in  $WW$ -cmf is zero, the invariant mass of the system is simply the total energy of all decay products:

$$M_{WW} = E_{\ell 1}^* + E_{\nu 1}^* + E_{\ell 2}^* + E_{\nu 1}^*. \quad (14)$$

- $\Delta\phi_{jj}$ :

When analyzing the Vector Boson Scattering (VBS), the leading and subleading tagging jets have a very specific kinematic signature: they are highly forward. Which means they have massive total energies and longitudinal momenta. The  $\gamma E\beta_T$  term becomes massive and so, the  $\Delta\phi_{jj}$  distribution is will be heavily distorted.

$$\vec{p}'_{Ti} = \vec{p}_{Ti} + \left[ \frac{\gamma - 1}{\beta^2} (\vec{\beta} \cdot \vec{p}_i) - \gamma E_i \right] \vec{\beta}_T. \quad (15)$$

### 2.2.3 Results

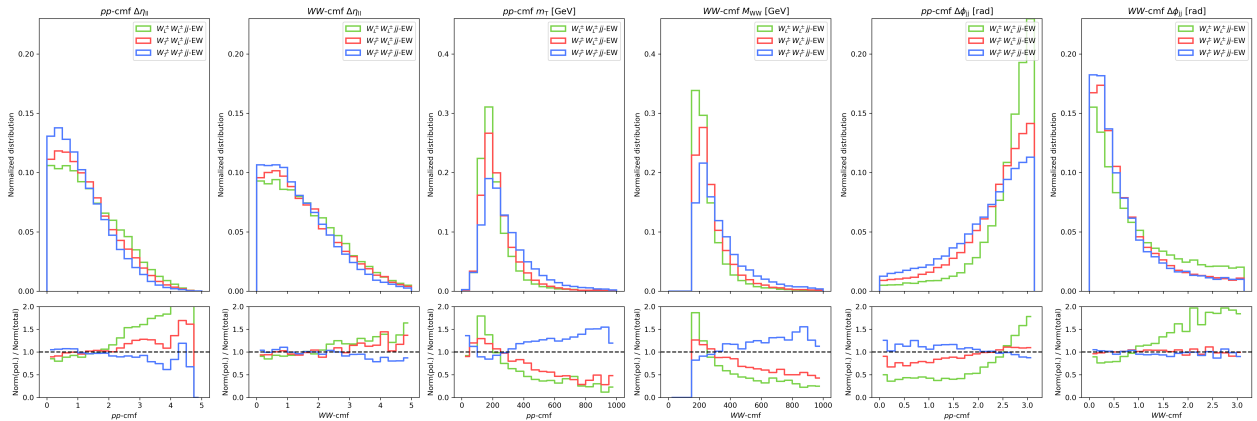


Figure 2: Distribution in the signal region after pre-cut and transformation to the  $WW$ -cmf in MadGraph for the difference in pseudorapidity between leading and subleading leptons, the transverse mass, and the difference in azimuthal angle between leading and subleading jets.

- **Peak values for  $\Delta\eta_{\ell\ell}$  decreases:** The decrease in the TT peak for  $\Delta\eta_{\ell\ell}$  in the  $WW$ -cmf occurs because boosting into the center-of-mass frame removes the global longitudinal “push” of the lab frame, unfolding the system and allowing the naturally broader angular distributions of transversely polarized bosons—which follow a  $(1 \pm \cos^2 \theta^*)$  distribution—to spread out and lower the normalized peak. Furthermore, because TT leptons are emitted more longitudinally, they are highly sensitive to “smearing” and potential under-boosting caused by the algebraic approximation of  $p_z$ , which further flattens the distribution compared to the LL state.
- **$M_{WW}$  goes to zero below  $\approx 160$  GeV:** The  $M_{WW}$  values drop to zero below approximately 160 GeV primarily due to the on-shell threshold; since we force a  $W$ -mass constraint of 80.385 GeV for each reconstructed boson, the total invariant mass cannot physically fall below the sum of those two masses ( $\sim 161$  GeV). Additionally, the selection criteria—specifically the high  $p_T$  requirements for leptons and the  $m_{jj} > 500$  GeV VBS topology cut—effectively filter out low-energy phase space where a lower  $M_{WW}$  might otherwise occur.
- **$\Delta\phi_{jj}$  goes to zero:** In  $pp$ -cmf, the  $\Delta\phi_{jj}$  distribution is governed by the hard scatter and recoil dynamics. But in  $WW$ -cmf, the distribution is heavily destroyed by the massive Energy term through the 3D boost. The jets are forced to point in similar transverse direction, making them highly collinear in  $\phi$ .



### 3 March 8th

#### 3.1 $WW$ -cmf in MadGraph

```
generate p p > w+{0} w+{0} j j QCD=0, w+ > l+ vl  
add process p p > w-{0} w-{0} j j QCD=0, w- > l- vl~  
output sample/script/LL  
launch sample/script/LL
```

```
shower = Pythia8  
detector = Delphes  
analysis= OFF  
madspin= OFF
```

```
set run_card nevents 100000  
set run_card ebeam1 6500  
set run_card ebeam2 6500
```

```
set run_card me_frame 3,4,5,6
```

```
set run_card cut_decays True  
set run_card ptj 34.0  
set run_card ptl 26.0  
set run_card miset 29.0
```

```
set run_card etaj 4.6  
set run_card etal 2.6
```

```
set run_card mmjj 450
```

```
set run_card ptj1min 64  
set run_card ptj2min 34
```

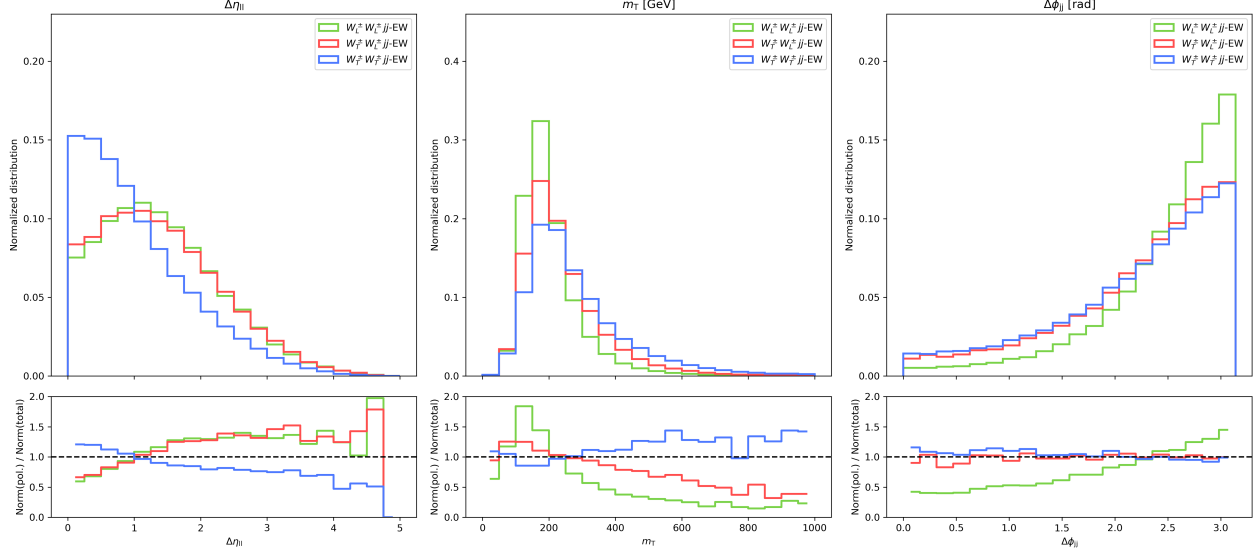


Figure 3: Distribution in the signal region after pre-cut in the  $WW$ -cmf in MadGraph for the difference in pseudorapidity, transverse mass, and difference in azimuthal angle.

|                   | $pp$ -cmf (fb) |       |       |       | $WW$ -cmf (fb) |       |       |       |
|-------------------|----------------|-------|-------|-------|----------------|-------|-------|-------|
|                   | LL             | LT    | TT    | Total | LL             | LT    | TT    | Total |
| Total             | 0.505          | 2.843 | 4.288 | 7.653 | 0.396          | 1.263 | 2.334 | 3.913 |
| Lepton cut        | 0.135          | 0.784 | 1.287 | 2.173 | 0.201          | 0.650 | 1.284 | 2.082 |
| MET cut           | 0.120          | 0.712 | 1.190 | 1.988 | 0.186          | 0.611 | 1.218 | 1.966 |
| Jet / VBS cut     | 0.096          | 0.586 | 0.959 | 1.616 | 0.148          | 0.484 | 0.960 | 1.547 |
| Jet cut (Thesis)  | 0.177          | 1.030 | 1.718 | 2.925 | 0.277          | 0.874 | 1.744 | 2.896 |
| Fraction          | 5.9%           | 35.7% | 58.4% | 100%  | 9.3%           | 30.4% | 60.3% | 100%  |
| Fraction (Thesis) | 6.1%           | 35.2% | 58.7% | 100%  | 9.6%           | 30.2% | 60.2% | 100%  |

Table 3: Comparison of cutflows, cross-sections, and polarization fractions for  $pp$ -cmf and  $WW$ -cmf frames. Results from local MadGraph generations are compared against the thesis references. All cross-sections are expressed in fb.

Table 3 compares the cutflow and polarization results of the local MadGraph generation against the reference thesis. While the absolute cross-sections in the signal region (Jet/VBS cut) are roughly 45% lower than the thesis values, the polarization fractions show excellent agreement. Specifically, the LL, LT, and TT fractions deviate by less than 1% from the reference in both the  $pp$ -cmf and  $WW$ -cmf frames. This confirms that despite the normalization difference, the underlying angular distributions and physics of the polarized states are correctly modeled. Figure 3 also shows the underlying kinematic distributions are similar to the results in the thesis.

## 3.2 Sherpa3

In Sherpa3, instead of generating an event that is 100 % purely "LL" or 100 % "TT", Sherpa generates a physical event (a specific arrangement of outgoing jets and leptons) and then calculates a weight for each polarization state for that exact same event.

Also, we must separate the  $W^+W^+jj$  and  $W^-W^-jj$  processes when using the Pol\_Cross\_Section feature because of how the generator strictly handles memory allocation for event weights.

### 3.2.1 Sample Preparation

For  $W^+$ :

```
EVENTS: 150000          #number of events to generate

BEAMS: [2212, 2212]     #incoming colliding particles; 2212 for a proton
BEAM_ENERGIES: 6500     #incoming particle energy for each

#specifies the internal structure model of the proton to be used during the collision
PDF_LIBRARY: LHAPDFSherpa      #tell Shepra to interface with the library
PDF_SET: PDF4LHC21_40_pdfas    #highly accurate PDF set

#93: any light quark, anti-quark, or gluon
#24: w+ boson
PROCESSES:
#2 incoming partons scatter to produce 2 same sign bosons and 2 outgoing partons
- 93 93 -> 24 24 93 93:
#forbidding any strong-force interactions in the hard matrix element
    Order: {QCD: 0, EW: 4}

SELECTORS:
#invariant mass of the 2 outgoing partons must be between 100-13,000 GeV
- [Mass, 93, 93, 100.0, 13000.0]

PARTICLE_DATA:
#treat w+ boson as strictly on-shell during the initial matrix element calculation
24:
    Width: 0.0
#turns on the decay module for heavy particles
HARD_DECAYS:
    Enabled: true
#crucial!!
#ensures the spin-correlations and polarization of w boson are calculated and passed
down to their decay products
    Pol_Cross_Section:
        Enabled: true
        Reference_System: COM      #COM:ww-cmf; Lab: pp-cmf
#disables specific decay routes for the w boson
#status : 0 meaning turns a decay channel off.
    Channels:
```

```

    '24,2,-1': {Status: 0}          #up+anti-down quark (hardronic decay)
    '24,4,-3': {Status: 0}          #charm+anti-strange quark (hardronic decay)
    '24,16,-15': {Status: 0}        #tau neutrino+anti-tau lepton (hardronic decay)
#dictates how and where the final simulated events are saved
EVENT_OUTPUT: HepMC3_GenEvent[Sherpa_Wp]

```

For  $W^-$ :

```
EVENTS: 150000
```

```
BEAMS: [2212, 2212]
```

```
BEAM_ENERGIES: 6500
```

```
PDF_LIBRARY: LHAPDFSherpa
```

```
PDF_SET: PDF4LHC21_40_pdfas
```

```
PROCESSES:
```

```

- 93 93 -> -24 -24 93 93:
  Order: {QCD: 0, EW: 4}

```

```
SELECTORS:
```

```
- [Mass, 93, 93, 100.0, 13000.0]
```

```
PARTICLE_DATA:
```

```

-24:
  Width: 0.0

```

```
HARD_DECAYS:
```

```
Enabled: true
```

```
Pol_Cross_Section:
```

```
Enabled: true
```

```
Reference_System: COM
```

```
Channels:
```

```

'-24,-2,1': {Status: 0}
'-24,-4,3': {Status: 0}
'-24,-16,15': {Status: 0}

```

```
EVENT_OUTPUT: HepMC3_GenEvent[Sherpa_Wm]
```

|            | Cross Sections (fb) |       |       |        | Event Pass Rates |          |
|------------|---------------------|-------|-------|--------|------------------|----------|
|            | LL                  | LT    | TT    | Unpol  | Raw Events       | Fraction |
| Total      | 1.239               | 3.781 | 6.874 | 11.915 | 299,998          | 1.0000   |
| Lepton cut | 0.577               | 1.710 | 3.230 | 5.479  | 142,376          | 0.4746   |
| MET cut    | 0.477               | 1.473 | 2.767 | 4.658  | 120,052          | 0.4002   |
| Jet cut    | 0.239               | 0.753 | 1.398 | 2.349  | 49,259           | 0.1642   |

Table 4: Cross sections for different polarization states and corresponding event pass rates at each selection cut stage.

According to table 4, the event pass rate for Sherpa simulation is much smaller than MadGraph due to less cuts in the event generating step. Perhaps we can explore whether there are other cuts can be added to Sherpa to enhance the event usage.

|                   | Sherpa $pp$ -cmf (fb) |       |       |        | Sherpa $WW$ -cmf (fb) |       |       |        |
|-------------------|-----------------------|-------|-------|--------|-----------------------|-------|-------|--------|
|                   | LL                    | LT    | TT    | Total  | LL                    | LT    | TT    | Total  |
| Total             | 1.060                 | 4.787 | 6.083 | 11.915 | 1.254                 | 3.816 | 6.873 | 11.915 |
| $n_l$ cut         | 0.472                 | 2.180 | 2.906 | 5.456  | 0.575                 | 1.714 | 3.214 | 5.456  |
| MET cut           | 0.381                 | 1.845 | 2.516 | 4.635  | 0.469                 | 1.470 | 2.755 | 4.635  |
| VBS cut           | 0.175                 | 0.915 | 1.306 | 2.335  | 0.237                 | 0.749 | 1.392 | 2.335  |
| VBS cut (Thesis)  | 0.182                 | 1.072 | 1.732 | 2.933  | 0.285                 | 0.909 | 1.778 | 2.933  |
| Fraction          | 7.3%                  | 38.2% | 54.5% | 100%   | 10.0%                 | 31.5% | 58.5% | 100%   |
| Fraction (Thesis) | 6.1%                  | 35.9% | 58.0% | 100%   | 9.6%                  | 30.6% | 59.8% | 100%   |

Table 5: Comparison of cutflows, cross-sections, and polarization fractions for  $pp$ -cmf and  $WW$ -cmf frames using Sherpa. Results from local generations are compared against the thesis Sherpa ( $W^\pm W^\pm jj$ ) references.

Table 5 shows that the calculated cross-sections are in close agreement with the thesis references. Furthermore, the polarization fractions for the LL, LT, and TT states remain consistent with the reference values across both the  $pp$ -cmf and  $WW$ -cmf frames.

## 4 March 12

### 4.1 Sherpa3 Rerun

It was identified that the initial analysis in section 3.2 lacked a proper detector simulation stage; specifically, Delphes was not applied to the Sherpa-generated events. Furthermore, the standalone Python implementation for jet reconstruction was unreliable for this analysis. Consequently, the Sherpa output files were re-processed through Delphes to ensure a more robust reconstruction of jets within the signal region.

Table 6 summarizes the results following this detector simulation. Notably, the discrepancy persists, which aligns with the results from MadGraph: the cross-sections obtained after the three signal region cuts are approximately 50% lower than those reported in the reference thesis. Despite this difference in absolute magnitude, the relative polarization fractions (LL, LT, and TT) remain broadly consistent with the thesis values.

|                   | Sherpa3 + Delphes $pp$ -cmf (fb) |       |       |        | Sherpa3 + Delphes $WW$ -cmf (fb) |       |       |        |
|-------------------|----------------------------------|-------|-------|--------|----------------------------------|-------|-------|--------|
|                   | LL                               | LT    | TT    | Total  | LL                               | LT    | TT    | Total  |
| Total             | 1.060                            | 4.787 | 6.083 | 11.915 | 1.254                            | 3.816 | 6.873 | 11.915 |
| $n_{ll}$ cut      | 0.265                            | 1.245 | 1.740 | 3.202  | 0.332                            | 0.985 | 1.905 | 3.202  |
| MET cut           | 0.217                            | 1.057 | 1.515 | 2.744  | 0.275                            | 0.847 | 1.645 | 2.744  |
| VBS cut           | 0.112                            | 0.585 | 0.870 | 1.540  | 0.154                            | 0.479 | 0.922 | 1.540  |
| VBS cut (Thesis)  | 0.182                            | 1.072 | 1.732 | 2.933  | 0.285                            | 0.909 | 1.778 | 2.933  |
| Fraction          | 7.1%                             | 37.4% | 55.5% | 100%   | 9.9%                             | 30.8% | 59.3% | 100%   |
| Fraction (Thesis) | 6.1%                             | 35.9% | 58.0% | 100%   | 9.6%                             | 30.6% | 59.8% | 100%   |

Table 6: Comparison of cutflows, cross-sections, and polarization fractions for  $pp$ -cmf and  $WW$ -cmf frames using Sherpa after Delphes detector simulation. Local generations are compared against the thesis Sherpa ( $W^\pm W^\pm jj$ ) references.

## 4.2 PDF Selection

I was using the preset settings for event generation in MadGraph of PDF selection:

```
nn23lo1 = pdlabel
230000 = lhaid
```

But to compare the results with Sherpa, we need to use the same PDF. Therefore, I decided to use the following settings:

In Sherpa3:

```
PDF_SET: PDF4LHC21_40_pdfas
```

In MadGraph:

```
set pdlabel lhapdf
set lhaid 93300
```

And I found that the cross section for different PDF changed:

|              | LL (fb) |         | LT (fb) |         | TT (fb) |         |
|--------------|---------|---------|---------|---------|---------|---------|
|              | PDF     | Nominal | PDF     | Nominal | PDF     | Nominal |
| Total        | 0.434   | 0.396   | 1.372   | 1.263   | 2.531   | 2.334   |
| $n_{ll}$ cut | 0.221   | 0.201   | 0.702   | 0.649   | 1.376   | 1.284   |
| MET cut      | 0.205   | 0.186   | 0.660   | 0.611   | 1.302   | 1.218   |
| VBS cut      | 0.163   | 0.148   | 0.518   | 0.484   | 1.017   | 0.960   |

Table 7: Comparison of cross-sections for different polarization states showing the effect of the PDF selection. The “PDF” columns represent the new PDF choice, while “Nominal” represents the baseline generation. All values are in fb.

To ensure consistency between MadGraph and Sherpa3, the default NNPDF2.3 LO (lhaid 230000) was replaced with the modern PDF4LHC21 benchmark (lhaid 93300).

As shown in table 7, updating to the PDF4LHC21 set results in a systematic increase in cross sections across all polarization states. This shift is expected, as the modern PDF set incorporated more precise experimental data and updated theoretical constraints compared to the decade-old nominal set. This transition ensures that the local simulations align with current LHC standards and provide more reliable results.

### 4.3 Particle-level phase space

After I read the thesis in detail again, I found out that the thesis was comparing the cross section at particle-level, not the detector-level. However, the previous cross section for my simulation were calculated at the detector-level. Instead of rerunning the event generations, we can use the free Delphes GenJet branch. Despite Delphes is a detector simulator, its default configuration actually runs the FastJet clustering algorithm on the PYTHIA truth particles before it applies any detector smearing.

- Particle-Level jets: Must run a jet clustering algorithm on all stable final state particles (Status = 1). (anti- $kt$  algorithm with  $R = 0.4$ ).
- Particle-Level MET: This is calculated as the negative vector sum of the transverse momenta of all stable, weakly interacting particles.
- Particle-Level Leptons: These are stable electrons and muons straight from the PYTHIA output.

```
particles = "Particle.PT", "Particle.Eta", "Particle.Phi", "Particle.Mass"...
genjets = "GenJet.PT", "GenJet.Eta", "GenJet.Phi", "GenJet.Mass"
genmet = "GenMissingET.MET"
```

| Sample                         | $WW$ -cmf         |                    |                    | $pp$ -cmf         |                    |                    |
|--------------------------------|-------------------|--------------------|--------------------|-------------------|--------------------|--------------------|
|                                | LL                | LT+TL              | TT                 | LL                | LT+TL              | TT                 |
| Thesis MadGraph                | 0.277 fb<br>9.58% | 0.874 fb<br>30.18% | 1.744 fb<br>60.24% | 0.177 fb<br>6.07% | 1.030 fb<br>35.20% | 1.718 fb<br>58.73% |
| Local MadGraph                 | 0.256 fb<br>9.59% | 0.814 fb<br>30.56% | 1.595 fb<br>59.85% | 0.166 fb<br>6.19% | 0.954 fb<br>35.60% | 1.560 fb<br>58.22% |
| Thesis Sherpa $W^\pm W^\pm jj$ | 0.285 fb<br>9.60% | 0.909 fb<br>30.59% | 1.778 fb<br>59.81% | 0.182 fb<br>6.09% | 1.072 fb<br>35.91% | 1.732 fb<br>58.00% |
| Local Sherpa3                  | 0.252 fb<br>9.87% | 0.780 fb<br>30.58% | 1.519 fb<br>59.54% | 0.192 fb<br>7.47% | 0.977 fb<br>37.93% | 1.406 fb<br>54.60% |

Table 8: Comparison of cross-sections and polarization fractions for  $WW$ -cmf and  $pp$ -cmf frames. The reference thesis values (MadGraph and Sherpa) are compared against the local MadGraph generations and local Sherpa3 generations at the particle-level VBS cut.

Table 8 provides a comprehensive comparison between local simulation and the reference thesis for both  $pp$ -cmf and  $WW$ -cmf. The polarization fractions show similar results. While the absolute cross section in the local generations are slightly lower than the thesis reference, which may likely due to higher order corrections or PDF sets. This result also shows that the underlying physics modeling and symmetry factors in MadGrpah or Sherpa3 setups are correctly aligned with the thesis results.



## 5 3/20

### 5.1 Background events

Following are the ranked background events mentioned in [2]:

1.  $W^\pm Z$  background (dominant)
2. Data-driven backgrounds ( $\sim 18$  % combined):
  - Non-prompt leptons: arises from leptons originating from hadron decays and jets that are misidentified as leptons. The source of these events is mainly  $W$ +jets and semi-leptonic  $t\bar{t}$  processes.
  - Charge-flip background (electron charge misidentification): This background primarily comes from  $Z \rightarrow e^+e^-$  decays. A high-energy electron emits a hard photon via bremsstrahlung, which then undergoes pair production into a high-energy positron and a soft electron. The soft electrons get lost in the detector, and the misidentified opposite-charged positron is measured instead, incorrectly passing the same-charge requirement.
3. QCD  $W^\pm W^\pm jj$  background ( $\sim 5.2$  %)
4. Small prompt backgrounds ( $< 2$  % combined)
5. Interference  $W^\pm W^\pm jj$ –INT ( $\sim 1.6$  %)

#### 5.1.1 $W^\pm Z jj$ background

- $pp \rightarrow W^\pm Z jj \rightarrow \ell^\pm \nu \ell^\pm \ell^\mp$  (including both pure electroweak  $W^\pm Z$ –EW and QCD-induced  $W^\pm Z$ –QCD scattering)
- Event generator:
  - $W^\pm Z$ –EW: Madgraph
  - $W^\pm Z$ –QCD: Sherpa2.2.12 + data-driven approach (will Shepra3 simulation be sufficient?) + control region restrictions
- CMF: evaluated in the  $WW$ -cmf and  $pp$ -cmf
- PDF set: NNPDF3.0\_nnlo\_as.0118

This background originates from the fully leptonic decay. It mimics the signal region criteria when one of the leptons from the  $Z$  boson decay fails baseline selection or falls completely outside the detector’s acceptance. This leaves exactly two same-charged leptons, two jets, and missing transverse momentum. It consists of both pure electroweak and QCD-induced scattering events.

Comprises the electroweak scattering  $W^\pm Z$ –EW and the strong scattering  $W^\pm Z$ –QCD.  $W^\pm Z$ –EW can be simulated by MadGraph well, while  $W^\pm Z$ –QCD is known to be poorly modeled by Monte Carlo simulations. Sample generated from Sherpa for  $W^\pm Z$ –QCD still needs to be corrected using the same data-driven approach as in [2].

### 5.1.2 QCD $W^\pm W^\pm jj$ background

- $pp \rightarrow \ell^\pm \nu \ell^\pm \nu jj$  via strong interaction vertices at order  $O(\alpha_{\text{EW}}^4 \alpha_S^2)$
- Event generator: MadGraph
- CMF: evaluated in the  $WW$ -cmf and  $pp$ -cmf
- PDF set: NNPDF3.0\_nnlo\_as\_0118

This is an irreducible background that yields the exact same final state particles as the signal, but it is produced at the order  $O(\alpha_{\text{EW}}^4 \alpha_S^2)$  involving strong QCD interaction vertices.

### 5.1.3 Interference background $W^\pm W^\pm jj$ –INT

- Quantum mechanical interference between the purely electroweak and QCD-induced  $W^\pm W^\pm jj$  diagrams that share identical external quarks, calculated at order  $O(\alpha_{\text{EW}}^5 \alpha_S)$
- Event generator: MadGraph and Sherpa

For MadGraph, the interference is calculated by:

$$N_{\text{int}} = N_{\text{unpol}} - \sum_{\text{pol}} N_{\text{pol}} \quad (16)$$

For Sherpa, the interference contribution is directly provided by an event weight as for the individual polarization states.

- CMF: evaluated in the  $WW$ -cmf and  $pp$ -cmf
- PDF set: modeled by MadGraph using NNPDF3.0

## 5.2 Control Region

### 5.2.1 $W^\pm Z$ control region

- Two signal leptons and an additional third signal lepton with  $p_T > 15$  GeV
- In the case of two electrons they have to fulfill  $|\eta| < 1.37$  and  $|m_{ee} - m_Z| > 15$  GeV
- At least two leptons satisfy  $m_{\ell\ell} \geq 20$  GeV
- $m_{\ell\ell\ell} \geq 106$  GeV
- $p_T^{\text{miss}} \geq 30$  GeV
- At least two signal jets with the highest  $p_T > 65$  GeV and the second highest  $p_T > 35$  GeV
- $m_{jj} \geq 200$  GeV
- $|\Delta y_{jj}| > 2$
- No jet is  $b$ -tagged using the DL1r tagger with the 85% efficiency working point

## 6 3/29

### 6.1 Signal events

Adjusting the PDF set for MadGraph into

```
set pdlabel lhpdf
set lhaid 260000
```

which sets directly the same PDF set used in the thesis (NNPDF30\_nlo\_as\_0118)

The following table is the after the correct PDF set selection with the thesis (NNPDF30\_nlo\_as\_0118). We can compare how different overlap removal affect the cross section. But it isn't clearly mentioned in the thesis how the overlap removal is applied.

| MadGraph  | with step 5 | without step 5 | corrected | thesis |
|-----------|-------------|----------------|-----------|--------|
| <b>LL</b> | 0.250       | 0.218          | 0.229     | 0.277  |
| <b>LT</b> | 0.794       | 0.662          | 0.699     | 0.874  |
| <b>TT</b> | 1.558       | 1.193          | 1.309     | 1.744  |

Table 9: Signal events generated from MadGraph with the PDF set: NNPDF30\_nlo\_as\_0118 and with different overlap removal conditions.

where the column “corrected” is set by the rules mentioned in the thesis 6.3.4 Fiducial Signal Region. If the resulting jet overlaps with an electron within  $\Delta R < 0.4$ , the electron is removed for  $p_{T,e}/p_{T,jet} < 0.5$ . The jet is removed if the electron carries more than 50% of the transverse jet momentum.

And the step 5 is a action to remove jets with  $R > 0.3$  of the signal leptons.

| Sherpa3   | with step 5 | without step 5 | corrected | thesis |
|-----------|-------------|----------------|-----------|--------|
| <b>LL</b> | 0.252       | 0.233          | 0.236     | 0.285  |
| <b>LT</b> | 0.780       | 0.700          | 0.717     | 0.909  |
| <b>TT</b> | 1.519       | 1.245          | 1.325     | 1.778  |

Table 10: Signal events generated from Sherpa3 with the PDF set: PDF4LHC21\_40\_pdfas and with different overlap removal conditions.

We can see from both table that different choice of the overlap removal affect the cross section very much. However, in both the thesis or in the references mentioned in the thesis didn't clearly denote what overlap removal is applied to the signal events. Therefore, the discrepancy between our simulation result and the result in the thesis might due to the choice of the overlap removal.

### 6.2 Background events

I found some tables and figures that we might be able to compare with the background events before training the DNN model. There are different kinds of cut mentioned in the thesis, which is summarized in the following figure: According to this figure 4, we can compare our simulation result with table G.2-5 in the thesis [3] which includes the number of signal and background events in SR, CR, and low  $m_{jj}$  region. Also, it includes both the  $pp$ -cmf and  $WW$ -cmf. Therefore, by the following equation,

$$N_{\text{expected}} = \sigma \times \mathcal{L} \times \epsilon \quad (17)$$

| Requirement                           | SR                                                                                 | Low- $m_{jj}$ CR         | WZ CR               |
|---------------------------------------|------------------------------------------------------------------------------------|--------------------------|---------------------|
| Leading and subleading lepton $p_T$   |                                                                                    | $> 27$ GeV               |                     |
| Electron $ \eta $                     | $< 2.47$ (1.37 in $ee$ ), excluding $1.37 \leq  \eta  \leq 1.52$                   |                          |                     |
| Muon $ \eta $                         |                                                                                    | $< 2.5$                  |                     |
| Leading (subleading) jet $p_T$        |                                                                                    | $> 65$ (35) GeV          |                     |
| Additional jet $p_T$                  |                                                                                    | $> 25$ GeV               |                     |
| Jet $ \eta $                          |                                                                                    | $< 4.5$                  |                     |
| $m_{\ell\ell}$                        |                                                                                    | $> 20$ GeV               |                     |
| $E_T^{\text{miss}}$                   |                                                                                    | $> 30$ GeV               |                     |
| Charge misid. $Z \rightarrow ee$ veto | $ m_{ee} - m_Z  > 15$ GeV                                                          |                          | –                   |
| $b$ -jet veto                         | $N_{b\text{-jet}} = 0, p_T^{b\text{-jet}} > 20$ GeV, $ \eta^{b\text{-jet}}  < 2.5$ |                          |                     |
| $N_{\text{veto leptons}}$             | $= 0$                                                                              | $= 0$                    | $= 1, p_T > 15$ GeV |
| $m_{\ell\ell\ell}$                    | –                                                                                  | –                        | $> 106$ GeV         |
| $m_{jj}$                              | $> 500$ GeV                                                                        | $200 < m_{jj} < 500$ GeV | $> 200$ GeV         |
| $ \Delta y_{jj} $                     |                                                                                    | $> 2$                    |                     |

Figure 4: Summary of the event selection for the signal region and control region.

- $N_{\text{expected}}$ : number of expected events
- $\sigma$ : the theoretical cross-section of the process
- $\mathcal{L}$ : the integrated luminosity of the data (for the full ATLAS Run 2 dataset, this is  $139 \text{ fb}^{-1}$ )
- $\epsilon$ : the efficiency and acceptance. (the event pass rate)

we might be able to validate our results before applying the DNN training.

## 7 4/10

### 7.1 Background simulation

#### 7.1.1 Simulation code

We generate the following background events by MadGraph 3.7.0 [4] and all of the PDF sets for each background simulation are selected to be `lhaid 260000`, which is `NNPDF30_nlo_as_0118`. For  $W^\pm W^\pm jj$  QCD

```
generate p p > l+ vl l+ vl j j QCD=2 QED=4
add process p p > l- vl~ l- vl~ j j QCD=2 QED=4
```

For  $W^\pm Z$  QCD

```
generate p p > l+ l- l+ vl j j QCD=0
add process p p > l+ l- l- vl~ j j QCD=0
```

For  $W^\pm Z$  EW

```
generate p p > l+ l- l+ vl j j QCD=2 QED=4
add process p p > l+ l- l- vl~ j j QCD=2 QED=4
```

### 7.1.2 Results

Table 11: Comparison of signal region cross sections (in fb) between simulation results, Stange Thesis [1] (Table G.2), and ATLAS paper [2] (Table 3). Expected yields from the thesis and paper have been converted to cross sections using the integrated luminosity mentioned in both paper of  $139 \text{ fb}^{-1}$ .

| Process                                                  | MadGraph3.7.1 [fb] | Sherpa3 [fb] | Thesis [1][fb] | Paper [2][fb] |
|----------------------------------------------------------|--------------------|--------------|----------------|---------------|
| <b>Signal Processes (<math>W^\pm W^\pm jj</math> EW)</b> |                    |              |                |               |
| $W_L W_L$ (LL)                                           | 0.132              |              | 0.145          | –             |
| $W_L W_T$ (LT)                                           | 0.426              |              | 0.453          | –             |
| $W_T W_T$ (TT)                                           | 0.847              |              | 0.922          | –             |
| sum                                                      | 1.405              | 1.353        | 1.520          | –             |
| $WW$ unpolarized                                         | 1.378              |              | 1.554          | 1.691         |
| <b>Background Processes</b>                              |                    |              |                |               |
| $WZ$ -EW                                                 | 0.150              | 0.409        | 0.108          | 0.107         |
| $WW$ -QCD                                                | 0.093              | 0.138        | 0.175          | 0.173         |
| $WZ$ -QCD                                                | 0.375              | 0.252        | 0.554          | 0.597         |

The signal region cuts obey the summarized table 2 in paper [2], but adding the detail that lepton pairs should be in same sign. We can see from table 11 that for electroweak processes (both signal and background), MadGraph [4] simulated a great result which is consistence with the reference. But for QCD-induced results, MadGraph [4] does not generate results reasonably. Therefore, we will try to generate some background events from Sherpa[5] to examine the consistency of the result. We can also see that for events generated by Sherpa[5], QCD-induced  $WW$  background behaves better than MadGraph, while MadGraph generates better for all electroweak events. However, there is still some inconsistency for the  $WZ$ -QCD induced background.

Table 12: Expected yields in the signal region (SR). The ATLAS results are shown alongside Thesis values and our simulation estimates using different generator versions, assuming an integrated luminosity of  $139 \text{ fb}^{-1}$ .

| Process              | MG 3.7.0 | Sherpa 3.0.3 | ATLAS | Thesis 9.1 | Thesis G.2 |
|----------------------|----------|--------------|-------|------------|------------|
| $W^\pm W^\pm jj$ EW  | 195.31   | 188.03       | 235   | 206.52     | 211.29     |
| $W^\pm W^\pm jj$ QCD | 12.95    | 19.12        | 24    | 24.05      | 24.27      |
| $W^\pm Z jj$ EW      | 20.81    | 56.98        | 14.9  | 14.95      | 15.07      |
| $W^\pm Z jj$ QCD     | 52.13    | 34.99        | 83    | 82.75      | 77.0       |

We can also compare the result by the expected yields in the signal region shown in table 12. Which shows the same trend that Sherpa generates better  $WW$ -QCD induced events than MadGraph, while MadGraph generates better electroweak processes ( $WWjj$  and  $WZjj$ ) than Sherpa.

However, both event generators exhibit low consistency for the  $WZjj$ -QCD induced process.

## 8 4/18

I took the following steps to try to generate results consistent with Feng-Yang's:

- **Cross section calculation for Sherpa:** Instead of extracting the nominal cross section from the `hepmc` file, we summed the cross sections for the positively charged processes (e.g.,  $W^+$ ) and the negatively charged processes (e.g.,  $W^-$ ), and then multiplied the total by the branching ratio.
- **Delphes `run_card`:** I initially used the ATLAS card but found that it lacked the fat jet finding configuration inside the `run_card` script. Therefore, I switched to the default `run_card` originally provided with Delphes and altered the jet radius to 0.4 to align with the paper.
- **Narrow width approximation (NWA) instead of the full process:** Because there were issues generating the full process in Sherpa (e.g., excessive integration time), we switched to using the narrow width approximation (NWA), which was also used for generating the signal process in the paper. Additionally, we adjusted the events generated by MadGraph to use NWA instead of the full process.

Table 13: Expected yields in the signal region (SR). The ATLAS results are shown alongside Thesis values and our simulation estimates using different generator versions, assuming an integrated luminosity of  $139 \text{ fb}^{-1}$ .

| Process              | MG 3.7.0 | Sherpa 3.0.3 | ATLAS | Thesis 9.1 | Thesis G.2 |
|----------------------|----------|--------------|-------|------------|------------|
| $W^\pm W^\pm jj$ EW  | 201.89   | 169.26       | 235   | 206.52     | 211.29     |
| $W^\pm W^\pm jj$ QCD | 15.20    | 17.59        | 24    | 24.05      | 24.27      |
| $W^\pm Zjj$ EW       | 21.94    | 32.05        | 14.9  | 14.95      | 15.07      |
| $W^\pm Zjj$ QCD      | 50.84    | 78.35        | 83    | 82.75      | 77.0       |

Comparing this table with last week's, we observe that for MadGraph, the  $WW$  processes are now closer to the reference values. For Sherpa, the expected yield for the  $WW$  processes has decreased, which might be related to the different Delphes `run_card` used. For the  $WZ$  processes, Sherpa demonstrates better consistency than last week's results.

To make further comparisons with the results from the thesis or paper, we need to determine the exact Delphes `run_card` that was used. As our latest results indicate, the choice of `run_card` significantly affects the cross section, as shown in the following table:

| Process      | Expected Events (139 fb <sup>-1</sup> ) |                        |
|--------------|-----------------------------------------|------------------------|
|              | delphes_card_default.tcl                | delphes_card_ATLAS.tcl |
| WWjj-EW-NWA  | 201.89                                  | 212.46                 |
| WWjj-QCD-NWA | 15.20                                   | 15.86                  |
| WZjj-EW-NWA  | 21.94                                   | 24.82                  |
| WZjj-QCD-NWA | 50.84                                   | 53.26                  |

Table 14: Comparison of expected event yields across four processes, evaluated using the ATLAS Delphes card versus the Default Delphes card.

## 9 4/26

### 9.1 Summary of DNN Training Input Variables

Table 15 summarizes the variables used as inputs for the neural network training. To improve the training process for the DNN algorithms, some variables are pre-scaled to achieve a more Gaussian-distributed input.

The  $\eta$  values of leptons and jets can be used to define the lepton Zeppenfeld variable:

$$Z_l^* = \left| \frac{\eta_l - \frac{1}{2}(\eta_{j1} + \eta_{j2})}{\eta_{j1} - \eta_{j2}} \right|. \quad (18)$$

This is a measure of how central a lepton is relative to the two tagging jets.

The definition of the massless early-projected transverse mass is:

$$m_{o1}^{ll,MET} = \sqrt{(p_T^{l1} + p_T^{l2} + p_T^{miss})^2 - |\vec{p}_T^{l1} + \vec{p}_T^{l2} + \vec{p}_T^{miss}|^2}, \quad (19)$$

This is an advanced kinematic approximation used to estimate system masses involving missing transverse momentum, assuming massless visible products.

And the mass-preserving transverse mass is defined by:

$$m_T^{ll,MET} = \sqrt{(E_T^{ll} + p_T^{miss})^2 - |\vec{p}_T^{l1} + \vec{p}_T^{l2} + \vec{p}_T^{miss}|^2}, \quad (20)$$

Or generally, it can be defined for object  $x$ , for  $x$  being the leading lepton, sub-leading lepton, or the de-lepton system:

$$m_T(x, E_T^{miss}) = \sqrt{(E_T^x + E_T^{miss})^2 - |\vec{p}_T^x + \vec{p}_T^{miss}|^2}, \quad (21)$$

where  $E_T = \sqrt{m^2 + |p_T|^2}$ .

### 9.2 Dataset separation

The training data is split into train validation, and test sets according to the event number in a ration of 3:1:1.

- The test set:  $((\text{EventNumber} - i_{\text{fold}}) \bmod 5) = 0$

| Kinematics                                                                              | Description                                                            | Scaling               |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------|
| <b>Low-level variables</b>                                                              |                                                                        |                       |
| $p_T^{\ell_1}, p_T^{\ell_2}, p_T^{j_1}, p_T^{j_2}$                                      | $p_T$ of the leading/subleading lepton/jet                             | $\log_{10}(x)$        |
| $\eta^{\ell_1}, \eta^{\ell_2}, \eta^{j_1}, \eta^{j_2}$                                  | $\eta$ of the leading/subleading lepton/jet                            | $x$                   |
| $\ell_1$ type, $\ell_2$ type                                                            | Flavor of the leading/subleading lepton                                | $x$                   |
| $\phi^{\ell_2} - \phi^{\ell_1}, \phi^{j_1} - \phi^{\ell_1}, \phi^{j_2} - \phi^{\ell_1}$ | Redefined $\phi$ of the subleading lepton/ leading jet/ subleading jet | $x$                   |
| $p_T^{\text{miss}}$                                                                     | Missing transverse momentum                                            | $\log_{10}(x)$        |
| $\phi(p_T^{\text{miss}}) - \phi^{\ell_1}$                                               | Redefined $\phi$ of the missing transverse energy                      | $x$                   |
| <b>High-level variables</b>                                                             |                                                                        |                       |
| $Z_{\ell_1}^*, Z_{\ell_2}^*$                                                            | Zeppenfeld variable of the leading/subleading lepton                   | $\sqrt{x}$            |
| $m_T^{\ell_1, \text{MET}}, m_T^{\ell_2, \text{MET}}$                                    | Transverse mass of the respective lepton and $p_T^{\text{miss}}$       | $\sqrt{x}$            |
| $\Delta R_{\ell\ell}, \Delta R_{jj}$                                                    | $\Delta R$ separation between the two leptons / two jets               | $x$                   |
| $\Delta\eta_{\ell\ell}$                                                                 | Pseudorapidity separation between the two leptons                      | $\sqrt{x}$            |
| $m_{\ell\ell}, m_{jj}$                                                                  | Invariant mass of the dilepton / dijet system                          | $\log_{10}(x)$        |
| $p_T^{\ell\ell}$                                                                        | Transverse momentum of the dilepton system                             | $\sqrt{x}$            |
| $m_T^{\ell\ell, \text{MET}}$                                                            | Transverse mass of the dilepton system                                 | $\sqrt{x}$            |
| $m_{o1}^{\ell\ell, \text{MET}}$                                                         | Early-projected massless invariant mass                                | $\sqrt{x}$            |
| $\Delta y_{jj}$                                                                         | Rapidity separation between the two jets                               | $x$                   |
| $\Delta\phi_{jj}$                                                                       | Azimuthal separation between the two jets                              | $x$                   |
| $(p_T^{\ell_1} \cdot p_T^{\ell_2}) / (p_T^{j_1} \cdot p_T^{j_2})$                       | $p_T$ ratio of the lepton and jet systems                              | $\log_{10}(x + 0.02)$ |
| $\min(\Delta R_{\ell_i, j_k})$                                                          | Minimal $\Delta R$ separation between any lepton and jet               | $x$                   |

Table 15: Summary of DNN input variables and their pre-scaling functions.



- The validation set:  $((\text{EventNumber} - i_{\text{fold}} + 1) \bmod 5) = 0$
- The training set: the remaining events

The index  $i_{\text{fold}}$  defines the different splits used for cross-validation in the hyperparameter optimization.

### 9.3 Training process

The training processes are divided into two parts: First, an inclusive network designed to separate the electroweak (EW) signal from all backgrounds. Second, the polarization-specific networks trained exclusively on the EW samples to distinguish specific polarization states. In details, it is trained to separate longitudinal single-boson polarization (LL and LT v.s. TT) or double-boson polarization (LL v.s. LT and TT) from the remaining transverse polarization states.

## 10 5/1

### 10.1 5-Fold Method

The  $k$ -fold method is a data-splitting and cross-validation strategy used to reliably evaluate neural network performance and prevent biases when making predictions.

#### 10.1.1 Data Categorization

The dataset is divided into three distinct and independent categories, using the  $i_{\text{Fold}}$  index:

- Test dataset: 20% with  $((\text{EventNumber} - i_{\text{Fold}}) \bmod 5) = 0$ . Kept completely hidden from the training and optimization process to test the final model.
- Validation dataset: 20% with  $((\text{EventNumber} - i_{\text{Fold}} + 1) \bmod 5) = 0$ . Used to evaluate the network's performance and tune hyperparameter during training.
- Training dataset: 60%, which is the remaining events that are not test or validation datasets. Used to adapt and optimize the network's trainable parameters.

where the index  $i_{\text{Fold}} \in [0, 1, 2, 3, 4]$  and independent of the physical properties of the events.

#### 10.1.2 Cross-Validation and Application

- Training Multiple Networks: The process iteratively trains five separate networks. Each network uses a different overlapping combination of the training folds and evaluates against non-overlapping validation and test sets.
- Evaluating Performance: Training across different independent validation datasets allows the author to calculate a mean network performance and estimate its statistical uncertainty.
- Preventing Mismodeling Bias: Neural networks naturally perform better on the data they were trained on than on unseen data. If predictions evaluated on training data were compared against measured data, it would introduce a systematic bias.
- Final Unbiased Predictions: To guarantee unbiased predictions across the entire dataset, each event's final prediction is generated exclusively by the specific DNN that held that event in its unseen test dataset.

### 10.1.3 Explanation of Proper Nouns

- **Unseen data** in the context of machine learning and the  $k$ -fold method refers specifically to the test dataset, which is used to provide an objective test of how well the final model can generalize to new, real-world examples. It is data that is kept completely hidden from the neural network while it is learning and optimizing its parameters.

$$s_j = \sum_i x_i w_{ij} + \theta_j \quad (22)$$

- **Weights** determine the strength and importance of the connection between two neurons. Mathematically, a weight is a multiplier. The output of a previous neuron ( $x_i$ ) is multiplied by its assigned weight ( $w_{ij}$ ) before it enters the next neuron.
- **Bias** ( $\theta_j$ ) is an additional parameter added to the neuron, independent of the input data. Mathematically, bias acts as an offset. By adding a bias, the network shifts the activation function left or right, determining how high the sum of the weighted inputs needs to be before the neuron activates.

## 10.2 Network Architecture

Based on section 9.2 in the reference thesis [3], the neural networks used are exclusively fully-connected feed-forward neural networks. The architecture is structured as follows:

- **Normalization Layer (Input):** Applies a scaling function to the input variables, normalizing their distributions to have a mean of 0 and a variance of 1. This ensures all inputs have similar scales without losing their kinematic phase space information.
- **Repeating Hidden Sequence:**

1. **Dense Layer:** A standard fully-connected layer.
2. **Batch Normalization Layer:** Used to normalize the output from the preceding dense layer.
3. **Activation Layer:** Applies the activation function, specifically the Swish function defined as:

$$f_{\text{Swish}}(x) = \frac{x}{1 + e^{-x}} \quad (23)$$

For a binary classification the Sigmoid function:

$$f_{\text{Sigmoid}}(x) = \frac{1}{1 + e^{-\beta x}} \quad (24)$$

4. **Dropout Layer:** Introduced for regularization to prevent overtraining.
- **Output Layer:** The repeating sequence concludes with a final dense layer, which computes the ultimate output score of the network.

## 11 5/10

### 11.1 Ap plying Simulated Data to Real Measured Data

Based on the thesis, there are three primary methods and strategies used to bridge the gap between Monte Carlo (MC) simulated training data and real measured data.

1. Preventing Mismodeling Bias via the  $k$ -Fold Method

Neural network naturally perform better on the data they were trained on than on unseen data. If a network’s prediction on its familiar training data was directly compared to real measured data, it would introduce an artificial systematic mismodeling (bias). The detailed explanation of this strategy is in Section 5.8 and Section 12.2.

To guarantee unbiased predictions across the entire dataset, each event’s final classification score is generated exclusively by the specific dense-NN that that held that event in its completely unseen “test” dataset.

2. Validation Strategy using Control Regions

Before deploying the neural network on the actual final measurement (the signal region), it is necessary to check if the simulation matches the real data. However, testing the dense-NN directly in the signal region could influence the final measurement and introduce a human “analysis strategy bias”. The definition of the control region is in Section 6.3.2 and the results of the validation tests are in Section 10.2.

Validation tests are safely performed in designated “control regions”. These are phase spaces that are kinematically similar to the signal region but are background-only. By comparing the dense-NN output on real measured data vs. simulated data in these control regions, we can ensure the simulation matches reality before looking at the actual signal.

\*Run trained DNN on the simulated background data and real measured data in control region and plot a histogram of the output scores (0(Background) to 1(Signal)). If it behaves correctly, two histograms should stack perfectly on top of each other and should assign mostly low scores to both datasets.

3. Correcting Simulation Errors with Data-Driven Backgrounds

Simulations are never perfect replicas of real life and fail to accurately predict certain detector artifacts, such as fake leptons or electrons flipping their charges. This approach is detailed in Section 7.2.2 and Section 7.2.3.

Since MC simulation models these backgrounds very poorly, the analysis uses data-driven techniques. This involves extracting metrics like “fake factor” directly from real-world measured data to overwrite or correct the flawed simulation predictions.

In summary, they first train five independent model on perfectly labeled simulated data. Then, prove the simulation matches reality in a safe control region and fix the broken parts (like fake-leptons)of the simulation using data-driven factors. Finally, the measured data is strictly evaluated by the specific model that corresponds to that fold. (If the real `EventNumber % 5` equals 2, that event is evaluated only by Model 2.)

### 11.2 References to Compare with

The thesis [3] contains specific figures that are highly valuable for validating our script, which is listed as follow:

### 1. Input Variable Distributions

In Appendix C, there are several figures of the pre-processed physical variables. We can plot these variables to verify the data parsing, variable calculations, and data shape match the thesis exactly.

### 2. Dense-NN Output Score Distributions

Once our model is trained, we can run it on the test dataset. We can plot our output scores using an equidistant binning of exactly 20 bins, as this is the standard binning convention used in the thesis for optimizing and visualizing Neural Network outputs. We can compare the distribution of the final classification probability scores outputted by the Sigmoid layer of our trained Dense-NN with Section 10.2.

### 3. ROC-AUC Performance Curves

We can compare our final testing AUC scores against the reported AUC scores in the thesis to confirm our network architecture and hyperparameters are performing at the expected efficiency, which can be found in Chapter 9.

## 11.3 ML Workflow

### 11.3.1 Data Pre-processing

Feng-Yang and me decided to use `parquet` as the dense-NN input variable file format. And the workflow should be as follows:

1. Extract Raw Data:
2. Calculate Variables: For each event, calculate all the low-level and high-level variables listed in Table 9.2 in the thesis [3].
3. Apply Pre-scaling: Apply the scaling mentioned in the thesis [3] in Table 9.2 to let the data resemble a Gaussian distribution for models learning more effectively.
4. Assign Folds: Calculate the physical Event Number and use the formula:

$$((\text{Event Number} - i_{\text{Fold}}) \bmod 5) = 0 \quad (25)$$

to tag each event with a fold number (0, 1, 2, 3, or 4).

5. Write to Parquet

## 12 5/17

### 12.1 Foundational ML Concepts

1. The **loss Curve** is a line graph that tracks how many “mistakes” the neural network is making overtime. In our script, the network uses Binary Cross-Entropy (BCE) loss, which is standard for binary classification. The lower the loss, the better the network is at distinguishing target from background.

2. An **epoch** is one complete cycle where the neural network has seen the entire training dataset exactly once. Because datasets in particle physics are massive, the data is fed to the network in smaller chunks called “batches.” (If we have 10,000 events and a batch size of 1,000, it takes 10 batches to get through all the data. Once all 10 batches have passes through the network, one epoch is complete.)

Reading it a second time (Epoch 2) helps understand and retain the material better, steadily lowering the loss.

3. **AUC** stands for area under the curve, and it specifically refers to the receiver operating characteristic (ROC) curve. The ROC-AUC evaluates the separation power of the model.

## 12.2 dense-NN Result

For the initial test, I am using 3592 events for the  $\text{DNN}_{W^\pm W^\pm}$  task and 6679 events for the  $\text{DNN}_{\text{pol}}$  task. The whole set of input features are used for the training. And the dense-NN structure for the two model is specified as follow:

```
history_ww = train_5_fold(
    df=df_ww,
    features_cols=features_cols,
    num_layers=8,
    neurons_per_layer=491,
    dropout_rate=0.355,
    lr=0.002,
    model_name="DNN_WW"
)

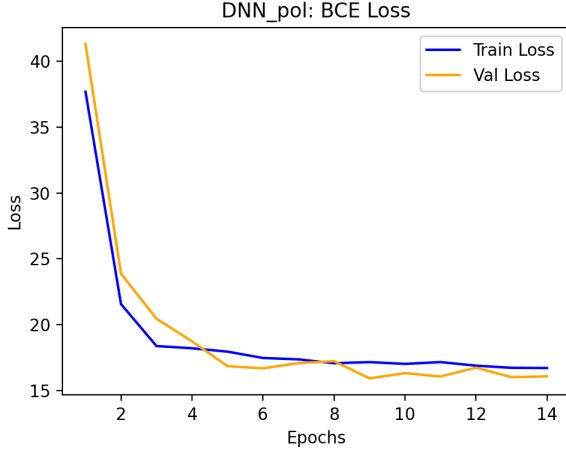
history_pol = train_5_fold(
    df=df_pol,
    features_cols=features_cols,
    num_layers=9,
    neurons_per_layer=296,
    dropout_rate=0.381,
    lr=0.0015,
    model_name="DNN_pol"
)
```

| Classification Task                   | Validation AUC      | Test AUC             |
|---------------------------------------|---------------------|----------------------|
| $\text{DNN}_{W^\pm W^\pm}$            | $0.7654 \pm 0.0107$ | $0.7601 \pm 0.0163$  |
| $\text{DNN}_{\text{pol}}$ (LL vs. TX) | $0.7878 \pm 0.0128$ | $0.7865 \pm 0.00083$ |

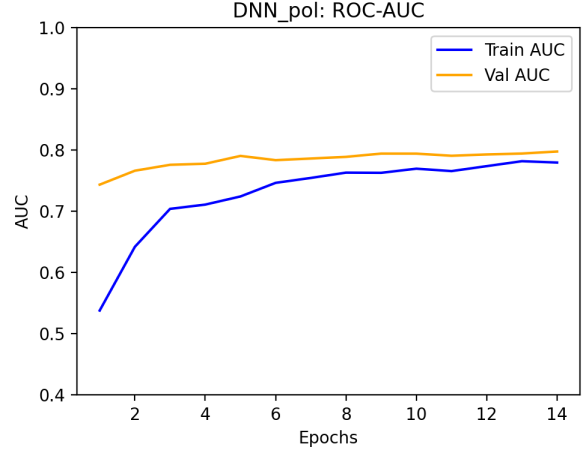
Table 16: Classification performance for DNN models using shared data.

Figure 5 illustrates the evolution of the loss and AUC curve. The quantitative performance of the models across the 5-fold ensemble is summarized in table 16.

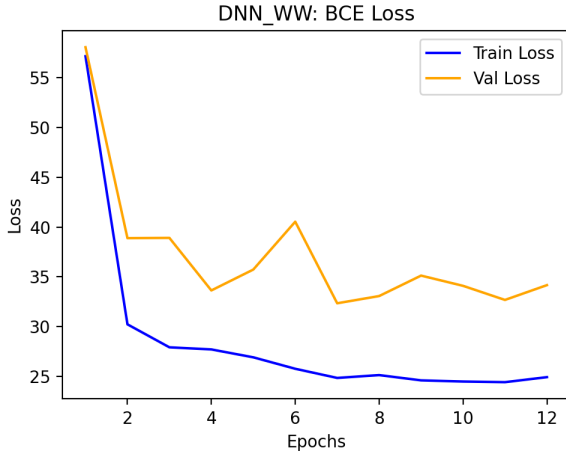
As anticipated for a preliminary test, the training curves indicate that the models—particularly  $\text{DNN}_{\text{WW}}$ —have not yet fully converged, as evidenced by the volatility in validation loss. The next steps will involve generating larger training samples to bridge the generalization gap, increasing the



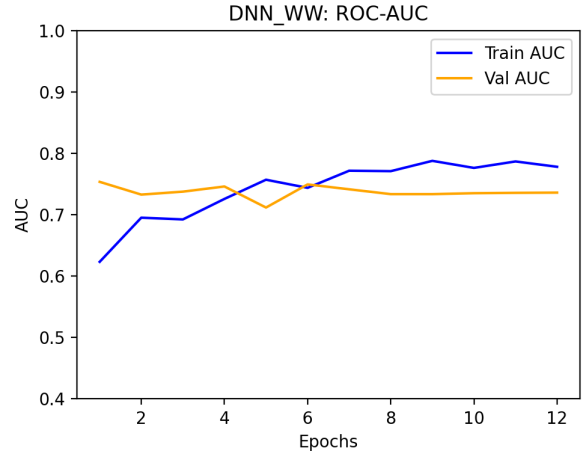
(a) DNN<sub>pol</sub>: BCE Loss



(b) DNN<sub>pol</sub>: ROC-AUC



(c) DNN<sub>WW</sub>: BCE Loss



(d) DNN<sub>WW</sub>: ROC-AUC

Figure 5: Model performance comparison: BCE Loss (left column) and ROC-AUC (right column) for DNN<sub>pol</sub> and DNN<sub>WW</sub>.

number of epochs to observe long-term stability, and testing optimized hyperparameters to improve the ROC-AUC plateau.

\*Table 9.1 in thesis [3] outlines the Class Weights. By multiplying the signal by a massive number so that the signal event looks just as loud and important as the background events.

## References

- [1] ATLAS Collaboration. “Evidence for longitudinally polarized  $W$  bosons in the electroweak production of same-sign  $W$  boson pairs in association with two jets in pp collisions at  $\sqrt{s} = 13$  TeV with the ATLAS detector”. In: *Phys. Rev. Lett.* 135 (2025), p. 111802. DOI: 10.1103/bp1n-ccq1.
- [2] The ATLAS Collaboration. “Measurement and interpretation of same-sign  $W$  boson pair production in association with two jets in  $pp$  collisions at  $\sqrt{s} = 13$  TeV with the ATLAS detector”. In: *JHEP* 04 (2024), p. 026. DOI: 10.1007/JHEP04(2024)026. arXiv: 2312.00420 [hep-ex].
- [3] Max V. Stange. “Longitudinal  $W^\pm$  Boson Polarization in Same-Charged  $W^\pm W^\pm$  Scattering in the ATLAS Experiment”. Doctoral thesis. Technische Universität Dresden, Jan. 2025.
- [4] J. Alwall et al. “The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations”. In: *JHEP* 07 (2014), p. 079. DOI: 10.1007/JHEP07(2014)079. arXiv: 1405.0301 [hep-ph].
- [5] Enrico Bothmann et al. “Event generation with Sherpa 3”. In: (Oct. 2024). arXiv: 2410.22148 [hep-ph].